From Text to Insight: Building a Semantic Question Answering System

AMS 691.02 Natural Language Processing
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Research Objective - Problem & Motivation:

Motivation:

Breaking Data Silos: Existing systems isolate religious texts by language, blocking the ability to search across traditions simultaneously.

Preserving Meaning: Sacred texts rely on deep abstract concepts that standard keyword search fails to capture faithfully.

Missing Benchmarks: No rigorous standard currently exists to determine which embedding model best handles spiritual semantics.

Objective:

Develop a **Unified Cross-Lingual RAG System** capable of answering complex queries across diverse theological corpora (Bhagavad Gita, Bible, Talmud).

Research Question:

"Which embedding architecture delivers the highest **semantic discriminability** and **retrieval accuracy** for a multi-scripture Retrieval-Augmented Generation (RAG) pipeline?"

Generative Layer: Model Selection & Rationale:

The Infrastructure Decision: Why Ollama?

- Leveraged **Ollama** to deploy a fully self-contained RAG pipeline.
- Ensures **zero data leakage**, eliminates external API costs, and guarantees reproducibility on local hardware.

The Comparative Analysis: Why Gemma 3?

- **Baseline:** Gemma 2 (2B) offered high speed but lacked nuance for complex theological synthesis.
- **Selected: Gemma 3 (4B)** emerged as the **optimal sweet spot**. It delivers superior reasoning capabilities and fluency required for abstract spiritual queries while remaining computationally feasible for our specific hardware constraints.

MODEL	SIZE (in billion parameter)	CONTEXT LENGTH	LANGUAGES	INPUT MODALITIES
Gemma 3 1B (IT)	1B	32k	English	Input: Text Output: Text
Gemma 3 4B (IT)	4B	128k	+140 Languages	Input: Text, Image Output: Text
Gemma 3 12B (IT)	12B	128k	+140 Languages	Input: Text, Image Output: Text
Gemma 3 27B (IT)	27B	128k	+140 Languages	Input: Text, Image Output: Text
Shield Gemma 2	4B	8k	+140 Languages	Input: Text, Image Output: Text

Embedding Model:

Qwen3-Embedding-0.6B:

- Modern multilingual semantic encoder designed for retrieval tasks.
- Good balance of quality vs. size for local CPU/MPS inference.
- Chosen to test whether a stronger encoder can significantly improve RAG over the same corpus.

Nomic-embed-text-v1.5:

- Widely used open embedding model for search and clustering.
- Provides a second strong, independent baseline (different family from Qwen) to check if gains are consistent.

Embeddinggemma:

- Runs fully via Ollama, no external API, easy to integrate.
- Represents a realistic deployment baseline we might actually ship, even if quality is unknown at the start.

Research Methodology:

No	Content
1	The Bhagavad Gita
2	CSB_Pew_Bible
3	The Talmud
4	Four-Vedas
5	Upanishads
6	Guru Granth Sahib Dhammapada
7	Mahabharata
8	Agama text
9	Thetanakh
10	17 Dhammapada

Data

How to Extract all Document Text

This script will take a document filename and generate a text file from all of its text.

The document can be any supported type.

The script works as a command line tool which expects the document filename supplied as a parameter. It generates one text file named "filename.txt" in the script directory. Text of pages is separated by a form feed character:

```
import sys, pathlib, pymupdf
fname = sys.argv[1]  # get document filename
with pymupdf.open(fname) as doc:  # open document
    text = chr(12).join([page.get_text() for page in doc])
# write as a binary file to support non-ASCII characters
pathlib.Path(fname + ".txt").write_bytes(text.encode())
```

The output will be plain text as it is coded in the document. No effort is made to prettify in any way. Specifically for PDF, this may mean output not in usual reading order, unexpected line breaks and so forth.

You have many options to rectify this – see chapter Appendix 2: Considerations on Embedded Files. Among them are:

1. Extract text in HTML format and store it as a HTML document, so it can be viewed in any browser.
2. Extract text as a list of text blocks via *Page.get_text("blocks")*. Each item of this list contains position information for its text, which can be used to establish a convenient reading order.
3. Extract a list of single words via *Page.get_text("words")*. Its items are words with position information. Use it to determine text contained in a given rectangle – see next section.

See the following two sections for examples and further explanations.

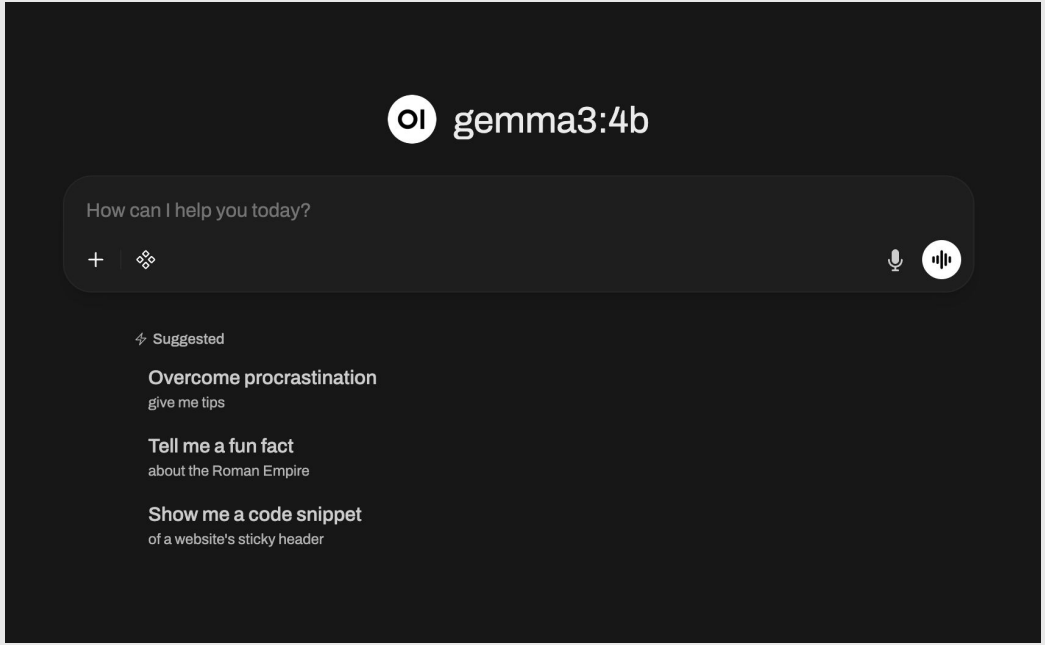
Page by Page text Extraction with PyMuPDF

Preprocessing:

1. Clean Noise
2. Chunk Size 600 Character with 100 character overlap
3. Metadata: Title, Chunk and Page

Data Preprocessing

Research Methodology:



embedded.py
Chunk PDF -> Embed each

embedded.py
Chunk -> store in Vector

pipelines.py
Query -> embed with ollama
nearVector search (current limit = 10) over Chunks

Deployment and Generate answer with Ollama gemma3:4b

Results and Analysis:

Evaluation 1: Contextual Relevance (DeepEval)

50 Gita-specific questions across:
Duty/karma, Atman, yoga paths, bhakti, detachment, Arjuna's doubt

For each model get top-k chunks by similarity, DeepEval scores how relevant retrieved context is to the question 20-question slice.

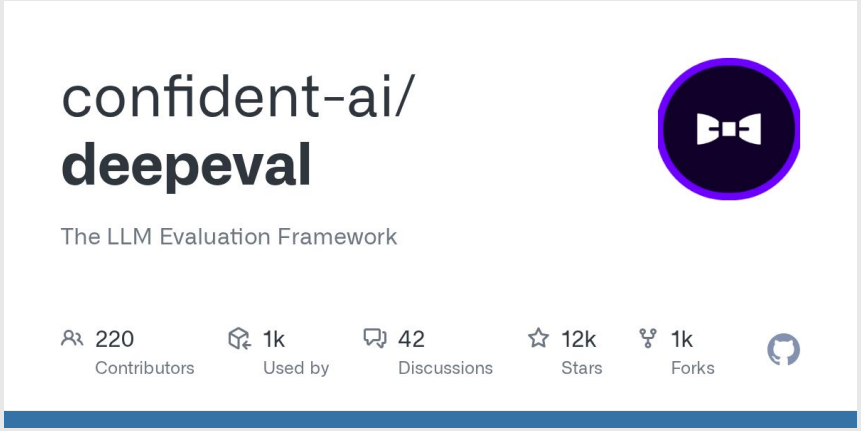
```
Question: What advice does Krishna give about remaining even-minded in success and failure?
Score: 0.714
Reason: The score is 0.71 because while the retrieval context includes relevant advice such as Krishna extolling t
he virtue of performing duties regardless of outcome and urging Arjuna not to yield to weakness, much of the conte
xt focuses on broader philosophical questions and future elaborations not directly addressing even-mindedness in s
uccess and failure, as noted in the reasons for irrelevancy.

Question: How does the Gita describe the right attitude toward success, failure, gain, and loss?
Score: 0.133
Reason: The score is 0.13 because most statements focus on the Gita's general background and content, which are ir
relevant to the attitude toward success, failure, gain, and loss, as noted in reasons like 'The Gita is one of the
 most widely read...' and 'The statement discusses metaphorical interpretations...'. However, relevant statements
such as 'Krishna extols the virtue of performing one's duties regardless of the outcome' and 'Krishna teaches Arju
na on the virtues of doing one's work' do address the right attitude, justifying the low but non-zero score.

Question: How does the Bhagavad Gita describe the nature of the self (Atman)?
Score: 0.038
Reason: The score is 0.04 because the reasons for irrelevancy highlight that most statements focus on qualities of
 nature, human traits, and chapters unrelated to the self (Atman), while only 'Chapter 13 - The Field and The Know
er of the Field' is relevant to the nature of the self as it discusses the distinction between the body and the se
lf.

Question: What does the Gita say about the immortality of the soul?
Score: 0.737
Reason: The score is 0.74 because while several statements like 'The soul is not subject to birth and death; it is
 eternal.' and 'Even when the physical body is destroyed, the soul does not perish.' directly address the Gita's t
eaching on the immortality of the soul, some parts of the retrieval context are irrelevant, such as 'The statement
 "This verse has origins in the Katha Upanishat..."' and 'The word 'vinasha' also means weakness...,' which do not
 pertain to the soul's immortality, lowering the overall relevancy.

Question: How does Krishna explain the difference between the body and the self?
Score: 0.667
Reason: The score is 0.67 because while many statements such as 'The charioteer (Krishna) and the passenger (Arjun
a) respectively represent the Supreme Soul and the individual soul resident in each body' and 'The individual soul
's intrinsic body is also indestructible' are relevant to explaining the difference between the body and the self,
 several reasons for irrelevancy highlight incomplete or unrelated information like 'The statement about the Lord'
s eternal and transcendental body is not directly related' and 'The statement discusses the reason for performing
duties as worship to the Lord, which is not directly about the difference between the body and the self.' This mix
 of relevant and irrelevant content justifies a moderate relevancy score.
```



Deepeval Score based on 20 question slice: Qwen3-Embedding-0.6B outperformed embedding gemma and nomic-embed-text-v1.5 with better performance of 100% and 4% respectively.

Interpretation: Qwen and Nomic both clearly outperform embedding gemma in QA

Evaluation 2: Multilingual MRR

What is the nature of the soul or self according to this scripture?

Okay, let's delve into the nature of the soul or self (Atman) as described in the Bhagavad Gita. It's a profoundly complex concept, and the Gita doesn't offer a simple, easily digestible answer. Here's a breakdown of the key ideas, categorized for clarity:

1. The Atman – Pure Consciousness (Shiva):

Eternal and Immutable: The Gita consistently portrays the Atman as eternal, unchanging, and beyond birth and death. It's not a temporary entity that fades away with the body. It's like pure consciousness, existing independently of the physical world.
The Basis of Reality: The Gita states, "There is only one being, the subject and the object, the seer and the seen, the doer and the deed." (Bhagavad Gita 13.23). The Atman is the fundamental reality underlying all appearances. It's the ground of being.
Non-Dual: The Atman is fundamentally beyond duality – good/bad, pleasure/pain, success/failure. These are all projections of the mind onto the Atman.

"आत्मा या स्वयं का स्वरूप इस शास्त्र के अनुसार क्या है?

Okay, translating that directly, "आत्मा या स्वयं का स्वरूप इस शास्त्र के अनुसार क्या है?" means "According to this scripture, what is the nature of the self (Atman)?"
Let's address that question directly, building on the previous explanation. Here's a detailed answer, mirroring the core teachings of the Bhagavad Gita:

According to the Bhagavad Gita, the Atman (the true self, soul, or Self) is:
Eternal and Immovable (Nitya & Achala): The Atman is not created nor destroyed. It is without beginning or end. It's fundamentally unchanging, unlike the body and mind, which are constantly changing. This is a crucial distinction.
Pure Consciousness (Shiva): It is not a physical entity or a "soul" in the sense of a separate, little person. It's pure, undifferentiated awareness—the underlying essence of all existence.

Identical with Brahman (Universal Consciousness): This is a key and often challenging concept. The Gita teaches that the Atman is ultimately identical with Brahman – the ultimate, universal consciousness, the ground of all being. It's like a drop of water is inseparable from the ocean.

¿Cuál es la naturaleza del alma o del yo según esta escritura?

Okay, let's address that question directly, translating it as "What is the nature of the soul or the self according to this scripture?" and providing a detailed response based on the Bhagavad Gita.

According to the Bhagavad Gita, the Atman (soul/self) is fundamentally:

Eternal and Unchanging (Sanatana & Achala): The Atman is utterly timeless. It doesn't experience birth, death, or any of the changes that occur in the phenomenal world.
Pure Consciousness (Shiva): It's not a localized entity or a "small soul." It's pure, undifferentiated awareness – the fundamental essence of being. Think of it like the vast, silent space that underlies all experiences.

The Source of Experience (The Knower): The Atman is the witness – the ultimate subject of all experience. Every thought, feeling, sensation, and perception arises within the Atman, but the Atman itself remains untouched by these experiences. It's the unchanging background of consciousness.
Identical with Brahman: This is a core concept. The Atman is ultimately identical to Brahman - the ultimate reality, the universal consciousness, the source of all existence.

Results and Analysis - MRR:

Example [edit]

Suppose we have the following three queries for a system that tries to translate English words to their plurals. In each case, the system makes three guesses, with the first one being the one it thinks is most likely correct:

Query	Proposed Results	Correct response	Rank	Reciprocal rank
cat	catten, cati, **cats**	cats	3	1/3
torus	torii, **tori**, toruses	tori	2	1/2
virus	**viruses**, virii, viri	viruses	1	1

Given those three samples, we could calculate the mean reciprocal rank as $(1/3 + 1/2 + 1)/3 = 11/18$, or approximately 0.61.

If none of the proposed results are correct, the reciprocal rank is 0.[1] Note that only the rank of the first relevant answer is considered, and possible further relevant answers are ignored. If users are also interested in further relevant items, mean average precision is a potential alternative metric.

```
=== Title discrimination for embedding model: nomic-embed-text-v1.5 ===
Question for 'The Bhagavad Gita' -> RR=1.000 (rank=1)
Question for 'The Bhagavad Gita' -> RR=1.000 (rank=1)
Question for 'CSB_Pew_Bible_2nd_Printing' -> RR=0.200 (rank=5)
Question for 'CSB_Pew_Bible_2nd_Printing' -> RR=0.000 (rank=>20)
Question for 'The Talmud' -> RR=0.000 (rank=>20)
Question for 'The Talmud' -> RR=0.000 (rank=>20)
Question for '17 Dhammapada (tr. Daw Mya Tin, by DPPS)_text' -> RR=1.000 (rank=1)
Question for '17 Dhammapada (tr. Daw Mya Tin, by DPPS)_text' -> RR=1.000 (rank=1)
Question for 'The-Upanishads-Translated-by-Swami-Paramananda' -> RR=1.000 (rank=1)
Question for 'The-Upanishads-Translated-by-Swami-Paramananda' -> RR=0.500 (rank=2)
Question for 'Siri Guru Granth - English Translation (matching pages)' -> RR=1.000 (rank=1)
Question for 'Siri Guru Granth - English Translation (matching pages)' -> RR=1.000 (rank=1)
Question for 'thetanakh' -> RR=0.143 (rank=7)
Question for 'thetanakh' -> RR=1.000 (rank=1)
Question for 'Four-Vedas-English-Translation' -> RR=0.059 (rank=17)
Question for 'Four-Vedas-English-Translation' -> RR=0.111 (rank=9)
Question for 'The Complete Mahabharata ' -> RR=0.100 (rank=10)
Question for 'The Complete Mahabharata ' -> RR=0.500 (rank=2)
Question for 'agama_023168_hr6' -> RR=0.000 (rank=>20)
Question for 'agama_023168_hr6' -> RR=0.000 (rank=>20)

MRR@20 for nomic-embed-text-v1.5: 0.481
Top-1 accuracy for nomic-embed-text-v1.5: 0.400
```

For Evaluation 2: Qwen3 outperformed both embeddinggemma and Nomic by 40% and 27% respectively.

```
Loading HuggingFace models on CPU (Qwen, nomic)...

=== Title discrimination for embedding model: Qwen3-Embedding-0.6B ===
Question for 'The Bhagavad Gita' -> RR=0.500 (rank=2)
Question for 'The Bhagavad Gita' -> RR=1.000 (rank=1)
Question for 'CSB_Pew_Bible_2nd_Printing' -> RR=1.000 (rank=1)
Question for 'CSB_Pew_Bible_2nd_Printing' -> RR=1.000 (rank=1)
Question for 'The Talmud' -> RR=0.000 (rank=>20)
Question for 'The Talmud' -> RR=0.000 (rank=>20)
Question for '17 Dhammapada (tr. Daw Mya Tin, by DPPS)_text' -> RR=1.000 (rank=1)
Question for '17 Dhammapada (tr. Daw Mya Tin, by DPPS)_text' -> RR=1.000 (rank=1)
Question for 'The-Upanishads-Translated-by-Swami-Paramananda' -> RR=1.000 (rank=1)
Question for 'The-Upanishads-Translated-by-Swami-Paramananda' -> RR=1.000 (rank=1)
Question for 'Siri Guru Granth - English Translation (matching pages)' -> RR=1.000 (rank=1)
Question for 'Siri Guru Granth - English Translation (matching pages)' -> RR=1.000 (rank=1)
Question for 'thetanakh' -> RR=0.250 (rank=4)
Question for 'thetanakh' -> RR=1.000 (rank=1)
Question for 'Four-Vedas-English-Translation' -> RR=0.250 (rank=4)
Question for 'Four-Vedas-English-Translation' -> RR=1.000 (rank=1)
Question for 'The Complete Mahabharata ' -> RR=0.500 (rank=2)
Question for 'The Complete Mahabharata ' -> RR=0.500 (rank=2)
Question for 'agama_023168_hr6' -> RR=0.000 (rank=>20)
Question for 'agama_023168_hr6' -> RR=0.000 (rank=>20)

MRR@20 for Qwen3-Embedding-0.6B: 0.650
Top-1 accuracy for Qwen3-Embedding-0.6B: 0.550
```

Evaluation 3: Title Discrimination across Scriptures: Can the embedding model pick the right book for each question?

20 questions total (2 per scripture):
Qwen MRR score was 0.654, Nomic MRR score was 0.561 and Embeddinggemma was 0.395

Evaluation 4: Qualitative Retrieval Examples: Example question: “What is the nature of the soul?”

Embeddinggemma: Mix of background and some relevant soul-related commentary
Qwen: Directly retrieves passage on verses 2.20–2.24, describing the soul as indestructible, without beginning or end
Nomic: Similar to Qwen, also surfaces detailed Atman descriptions

Evaluation 5: Retrieval Coverage across Books: “Which scripture dominates retrieval for generic spiritual questions?”

For all 3 models, top 40 hits are 100% from The Bhagavad Gita Reason: Gita is dense and directly matches question wording, so it dominates similarity Confirms that question phrasing + document content strongly.

Demo:

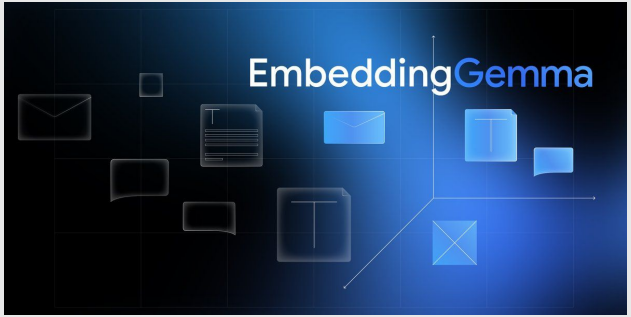
gemma3:4b
Set as default
AK
gemma3:4b
How can I help you today?
Suggested
Explain options trading
if I'm familiar with buying and selling stocks
Show me a code snippet
of a website's sticky header
Overcome procrastination
give me tips

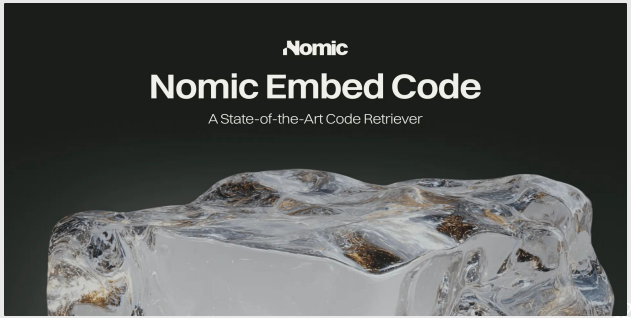
Conclusion:

- Qwen3 Embedding 0.6B delivers the strongest performance for multi scripture retrieval
- Nomic embed text v1.5 is a solid alternative with near parity
- Embeddinggemma is viable for fully local deployments despite weaker scores
- The evaluation pipeline offers a repeatable way to benchmark models in other domains

Future work:

- Reindex Weaviate with Qwen or nomic vectors for live system
- Add hybrid retrieval (BM25 + vector) and per-title filters Expand eval set with more languages.
- Investigate answer faithfulness with dedicated metrics





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Thank You!

Github Link: